MONASH University

STATISTICS

Session 6: Hypothesis Testing

Observing Chance

We often hear people report astonishing findings:

- A remarking coincidence
- An unusually strong or weak effect
- A result that seems far too unlikely
- An unusual sequence of events

When we hear something astonishing, our first instinct may be to ask

- “How could this possibly happen?”

But in statistics, there is another question we need to ask first:

- “Is this result truly astonishing, or could we have seen something this extreme simply by random chance?”

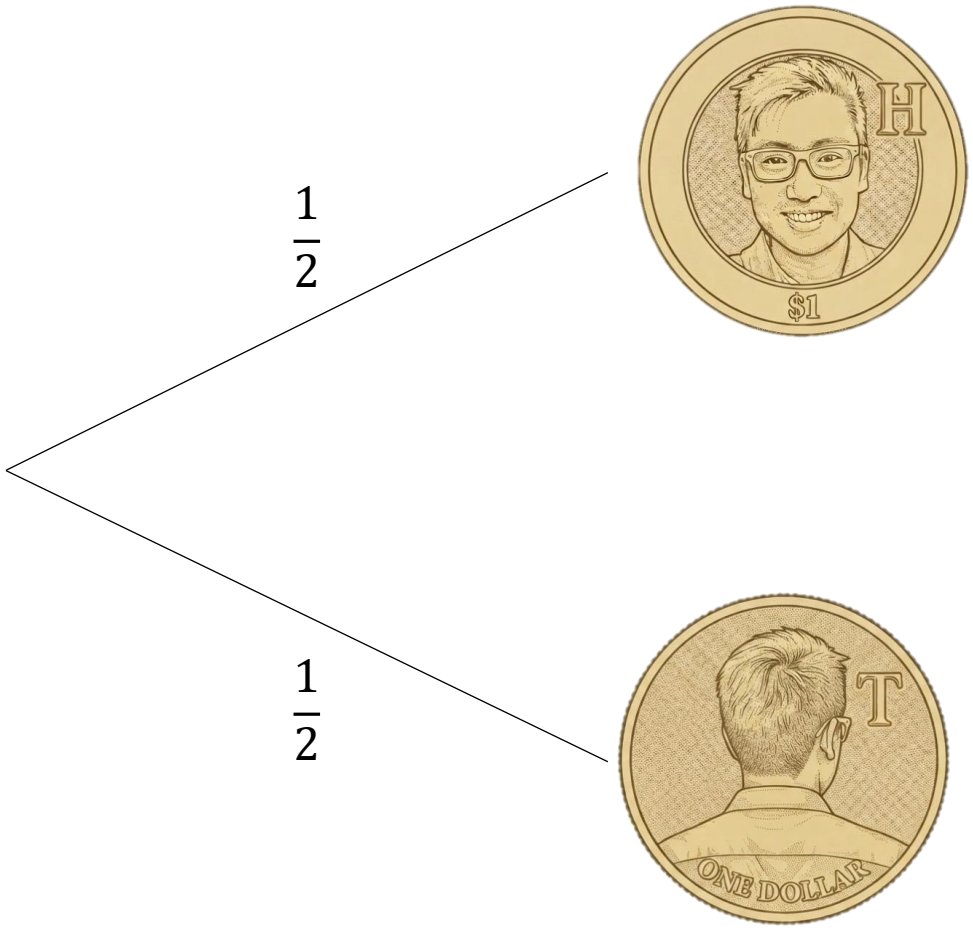
Observing Chance

- Suppose I tell you that “I’m psychic and can see the future.”
- To prove my claim (that I am psychic), I flip a coin then cover it.
- Would you be impressed is I correct called the outcome of the coin flip?
  - Probably not, because my ‘prediction’ had a 50% chance of being correct by guessing alone



MONASH University

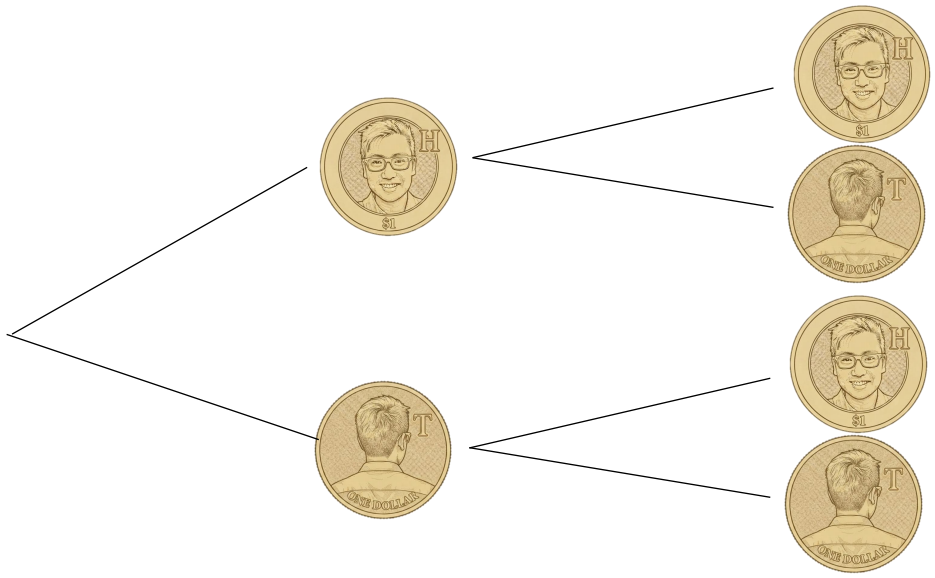
Observing Chance



Minh is psychic	VS	The results were due to random chance
Based on 1 flip		50%

MONASH University

Observing Chance



Minh is psychic

VS

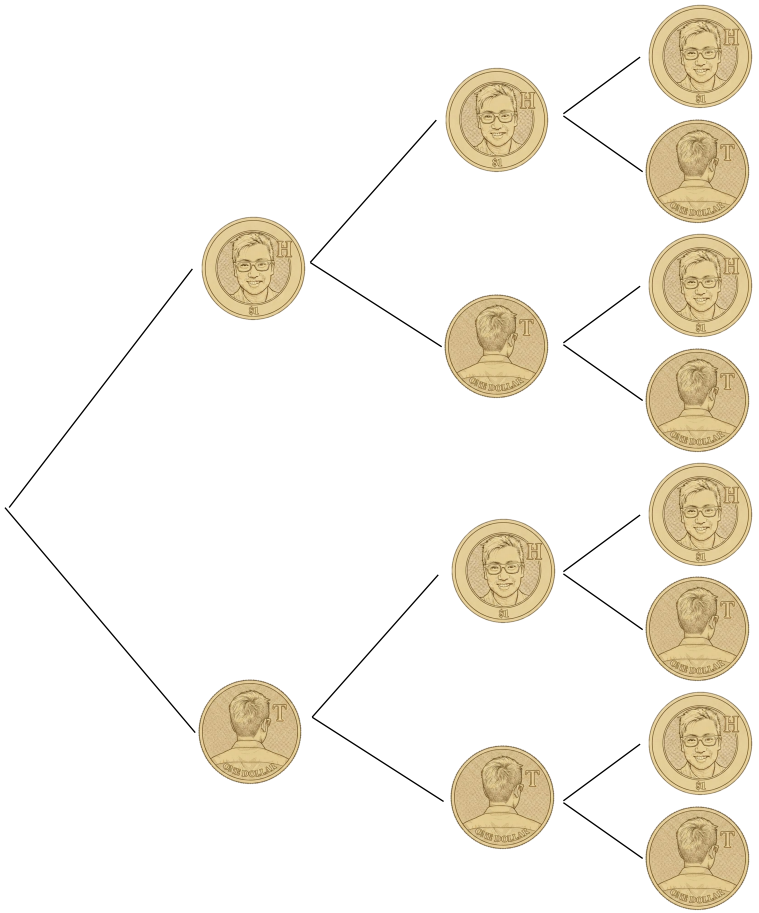
The results were due to random chance

Based on 2 flips

25%

MONASH University

Observing Chance



Minh is psychic

Based on 3 flips

VS

The results were due to random chance

12.5%

MONASH University

Observing Chance

"Is this result truly astonishing, or could we have seen something this extreme simply by random chance?"

The Lady Tasting Tea



	Cup 1	Cup 2	Cup 3	Cup 4	Cup 5	Cup 6	Cup 7	Cup 8
Decision:	M or T	M or T	M or T	M or T	M or T	M or T	M or T	M or T

MONASH University

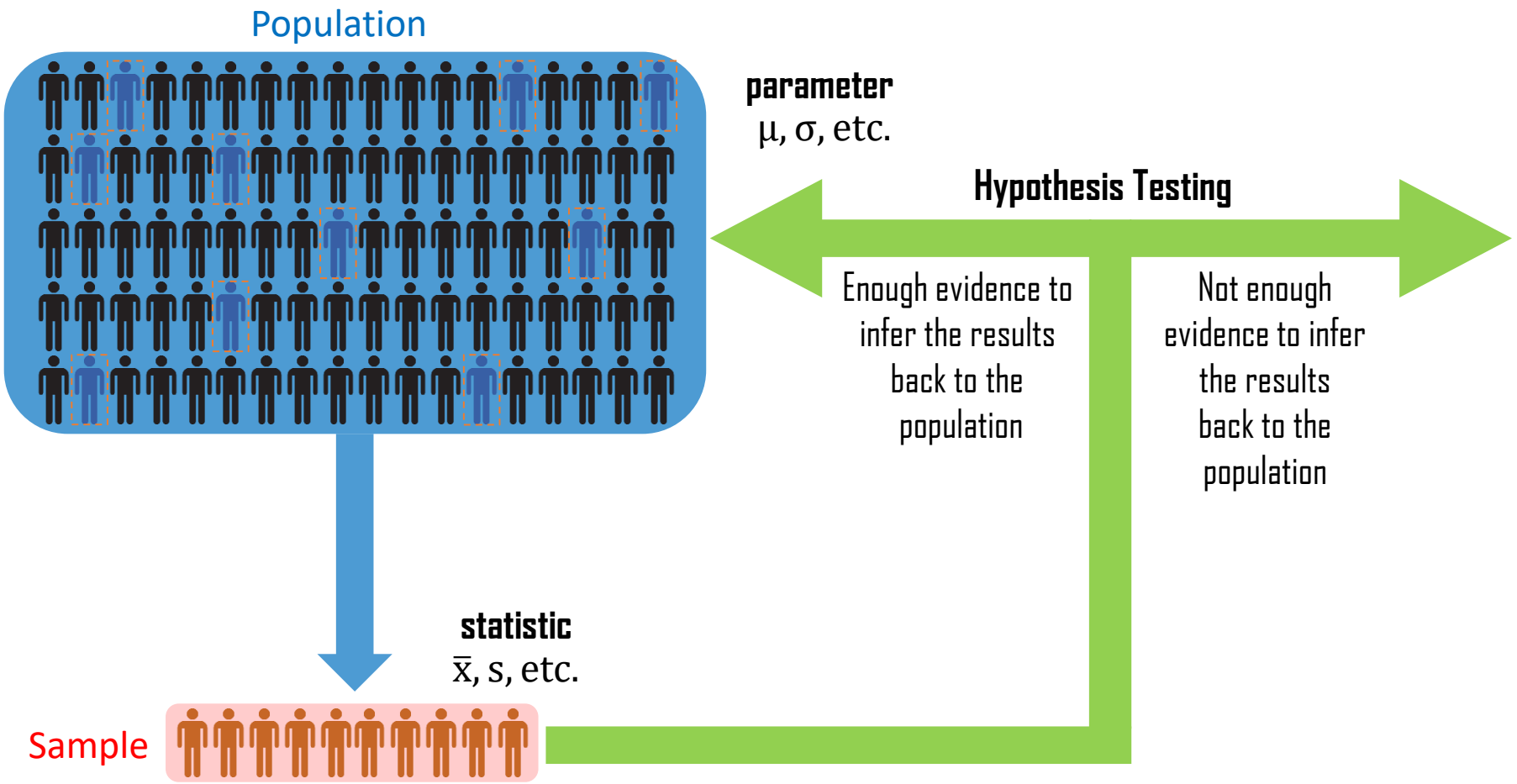
The Lady Tasting Tea

	Cup 1	Cup 2	Cup 3	Cup 4	Cup 5	Cup 6	Cup 7	Cup 8
Combination 1	M	M	M	M	T	T	T	T
Combination 2	M	M	M	T	M	T	T	T
Combination 3	M	M	M	T	T	M	T	T
Combination 4	M	M	M	T	T	T	M	T
Combination 5	M	M	M	T	T	T	T	M
Combination 6	M	M	T	M	M	T	T	T
Combination 7	M	M	T	M	T	M	T	T
Combination 8	M	M	T	M	T	T	M	T
Combination 9	M	M	T	M	T	T	T	M
Combination 10	M	M	T	T	M	M	T	T
Combination 11	M	M	T	T	M	T	M	T
Combination 12	M	M	T	T	M	T	T	M
Combination 13	M	M	T	T	T	M	M	T
Combination 14	M	M	T	T	T	M	T	M
Combination 15	M	M	T	T	T	T	M	M
Combination 16	M	T	M	M	M	T	T	T
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
Combination 70	T	T	T	T	M	M	M	M

$$\frac{1}{70} = 0.014$$

MONASH University

Hypothesis Testing



Hypothesis Testing

Everyday questions	Hypothesis-Testing questions
"Do Australians earn a good income?"	"Is the mean annual income of Australian adults different from $100,000?"
"Does studying more improve exam results?"	"Do students who study at least 10 hours per week have a higher mean exam score than students who study less than 10 hours?"
"Does studying more improve exam results?"	"Is there a positive relationship between the number of hours studied and exam score?

Null Hypothesis Significance Testing (NHST)

A classical framework used to decide whether observed data provide enough evidence to reject a default assumption (usually of no effect or no difference)

Null Hypothesis (H_0)

The default or baseline position in a hypothesis test. It typically states that there is no effect, no difference, or no relationship in the population. It provides a benchmark against which we evaluate the evidence in our sample data.

Alternative Hypothesis (H_1)

The claim or effect that we are looking for evidence in favour of. It typically states that there is an effect, difference, or relationship in the population, or that a population parameter differs from a specified value.

Null and Alternative Hypotheses

Everyday questions	Hypothesis-Testing questions	Null and Alternative hypotheses
"Do Australians earn a good income?"	"Is the mean annual income of Australian adults different from $100,000?"	$H_0: \mu = 100{,}000$ $H_1: \mu \neq 100{,}000$
"Does studying more improve exam results?"	"Do students who study at least 10 hours per week have a higher mean exam score than students who study less than 10 hours?"	$H_0: \mu_{\geq 10} = \mu_{<10}$ $H_1: \mu_{\geq 10} > \mu_{<10}$
"Does studying more improve exam results?"	"Is there a positive relationship between the number of hours studied and exam score?	$H_0: \beta = 0$ $H_1: \beta > 0$

MONASH University

The Role of the Null Hypothesis

It is the position we assume for the purpose of the test. We then ask whether the data look unusual under that assumption, and if so, we can reject the null.

Null and Alternative Hypotheses

Hypothesis-Testing questions	Null hypothesis	Alternative hypothesis
"Is the mean annual income of Australian adults different from $100,000?"	$H_0: \mu = 100{,}000$	$H_1: \mu \neq 100{,}000$
"Do students who study at least 10 hours per week have a higher mean exam score than students who study less than 10 hours?"	$H_0: \mu_{\geq 10} = \mu_{<10}$	$H_1: \mu_{\geq 10} > \mu_{<10}$
"Is there a positive relationship between the number of hours studied and exam score?	$H_0: \beta = 0$	$H_1: \beta > 0$

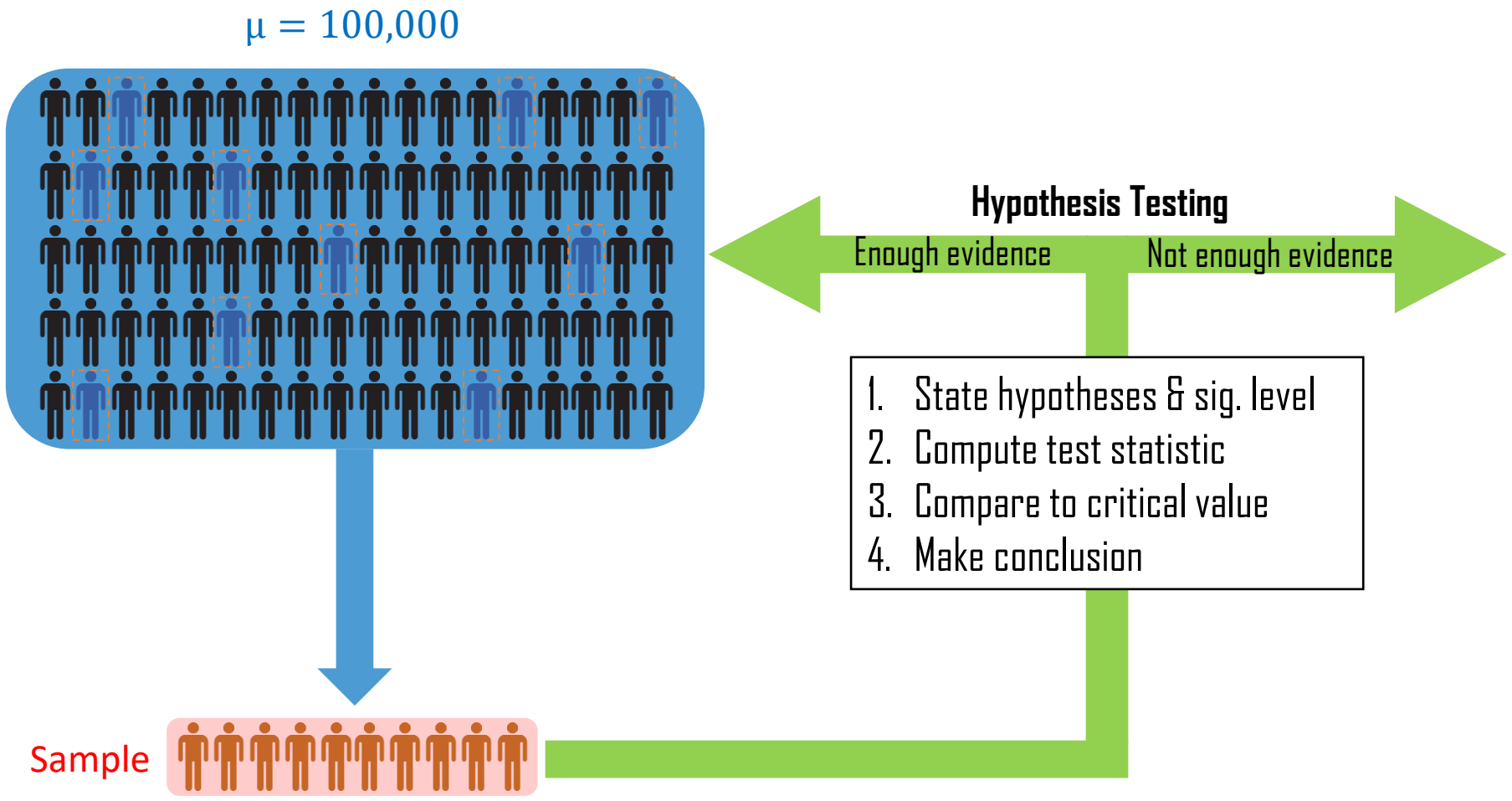
MONASH University

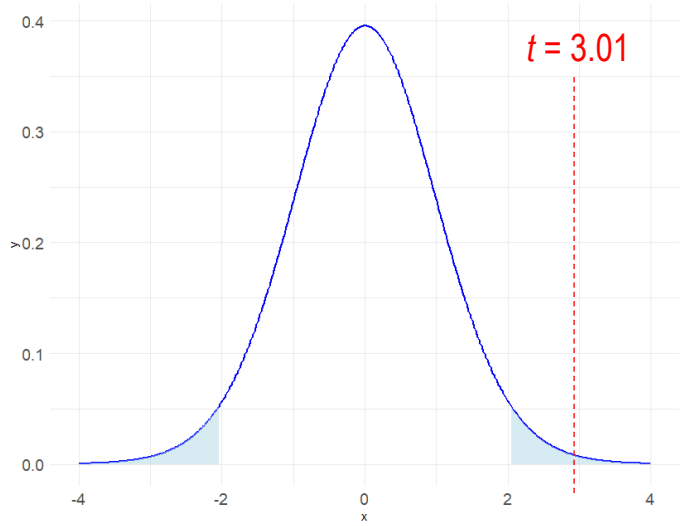
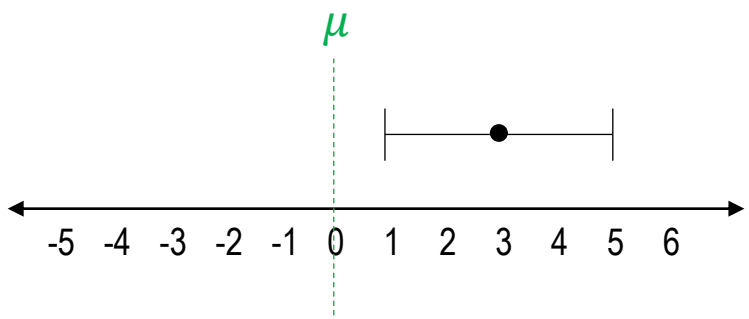
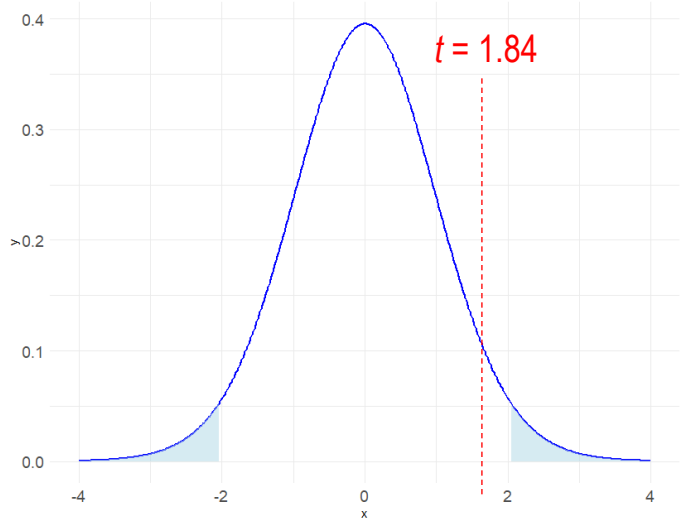
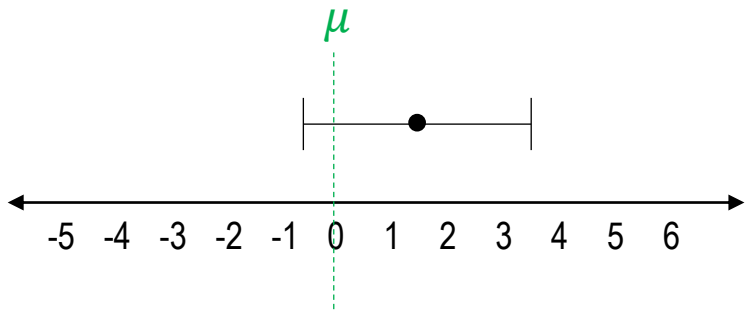
Directional Hypotheses



MONASH University

Test Statistic



	Critical value method	p-value method	Confidence Interval method
Reject H_0 Statistically significant	When the test statistic (t) is more extreme than the critical region 	When the p-value for the computed test statistic is less than the significance level* $p < \alpha$ * Unless otherwise stated, use $\alpha = 0.05$	When the confidence interval does not capture μ 
Fail to reject H_0 Not Statistically significant	When the test statistic (t) is not more extreme than the critical region 	When the p-value for the computed test statistic is greater than the significance level* $p > \alpha$ * Unless otherwise stated, use $\alpha = 0.05$	When the confidence interval does capture μ 

Critical Value Method

Suppose a recent report suggested that the mean annual income of Australians is $100,000. You want to test this claim so you sample a small number of households (n = 30) and find that the mean and std. deviation of your sample are $108,000 and $15,000, respectively.

Using the NHST framework, determine if your estimate is significantly different to the claimed value, or if the results of your sample are due to chance.

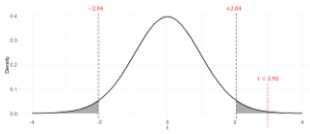
Step 1. State hypotheses and significance level

$$H_0: \mu = 100{,}000 \qquad H_1: \mu \neq 100{,}000 \qquad \alpha = 0.05$$

Step 2. Compute the test-statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{108000 - 100000}{15000/\sqrt{30}} \approx 2.92$$

Step 3. Compare test-statistic to critical value



$$t = 2.92 > t^* = 2.04$$

Step 4. Make a conclusion

- Reject H_0
- There is statistically significant evidence that the mean annual income of Australian households differs from $100,000.

MONASH University

P-value Method

Suppose a recent report suggested that the mean annual income of Australians is $100,000. You want to test this claim so you sample a small number of households (n = 30) and find that the mean and std. deviation of your sample are $108,000 and $15,000, respectively.

Using the NHST framework, determine if your estimate is significantly different to the claimed value, or if the results of your sample are due to chance.

Step 1. State hypotheses and significance level

$$H_0: \mu = 100{,}000 \qquad H_1: \mu \neq 100{,}000 \qquad \alpha = 0.05$$

Step 2. Compute the test-statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{108000 - 100000}{15000/\sqrt{30}} \approx 2.92$$

Step 3. Compute the p-value associated with the test statistic and compare to the significance level

$$p = 0.0067 < \alpha = 0.0500$$

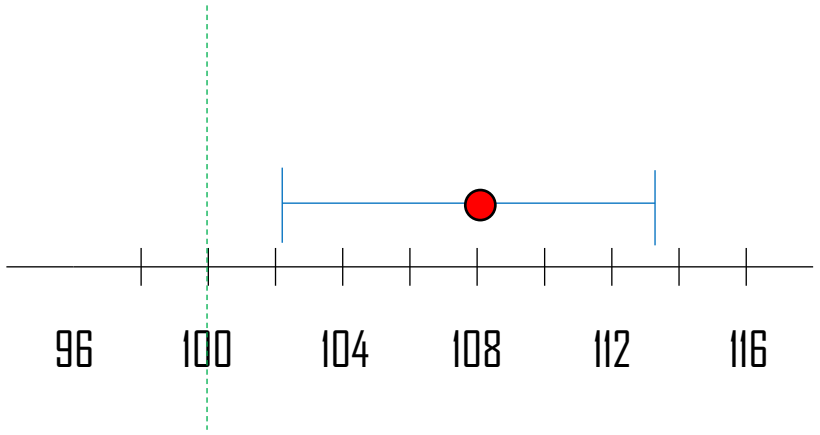
Step 4. Make a conclusion

- Reject H_0
- There is statistically significant evidence that the mean annual income of Australian households differs from $100,000.

MONASH University

Confidence Interval Method

Suppose a recent report suggested that the mean annual income of Australians is $100,000. You want to test this claim so you sample a small number of households (n = 30) and find that the mean and std. deviation of your sample are $108,000 and $15,000, respectively.



MONASH University

Step 1. State hypotheses and significance level

$H_0: \mu = 100{,}000$ $H_1: \mu \neq 100{,}000$ $\alpha = 0.05$

Step 2. Compute the test-statistic

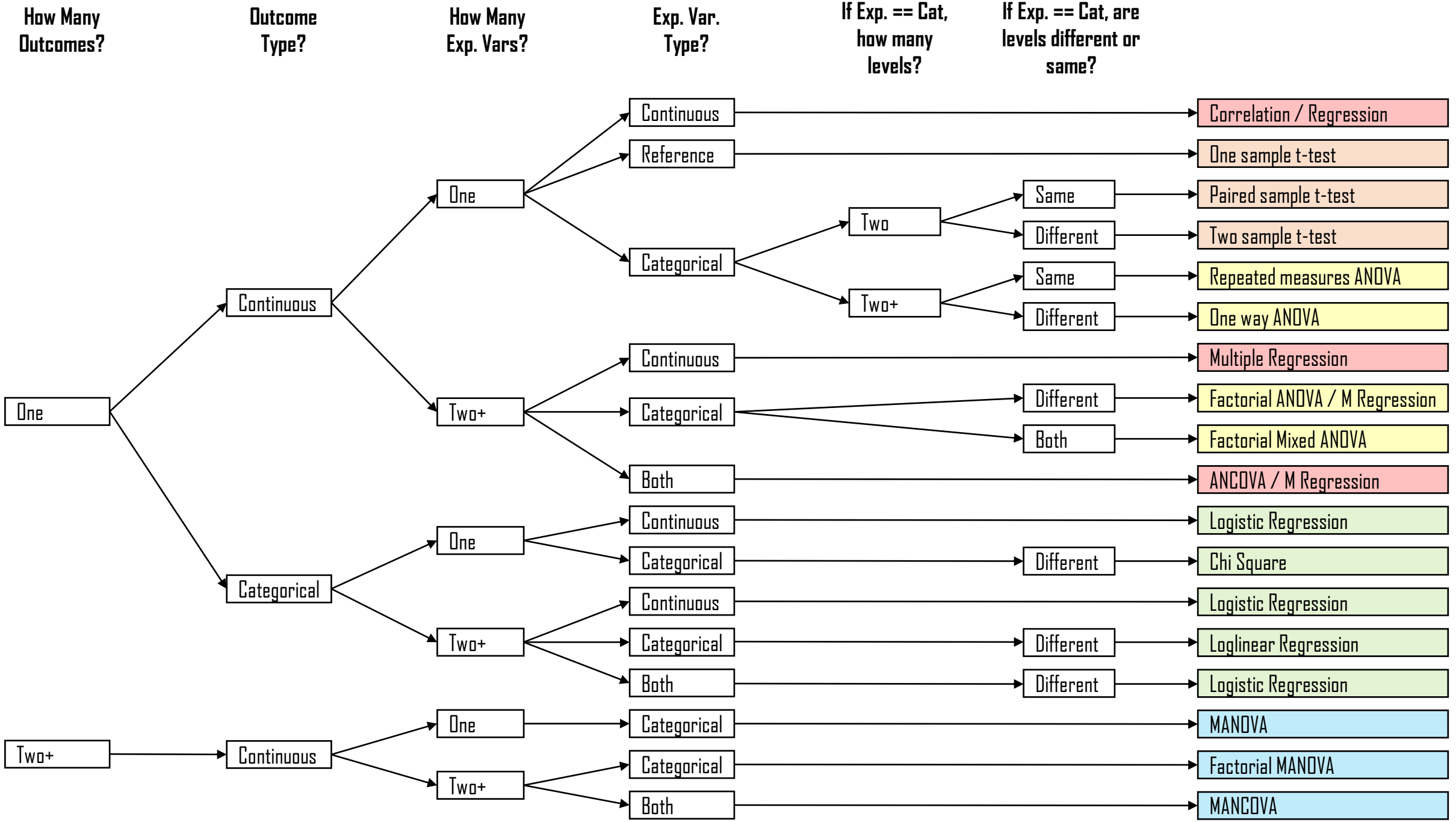
$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{108000 - 100000}{15000/\sqrt{30}} \approx 2.92$$

Step 3. Compute the CI and see if it capture the population parameter of interest

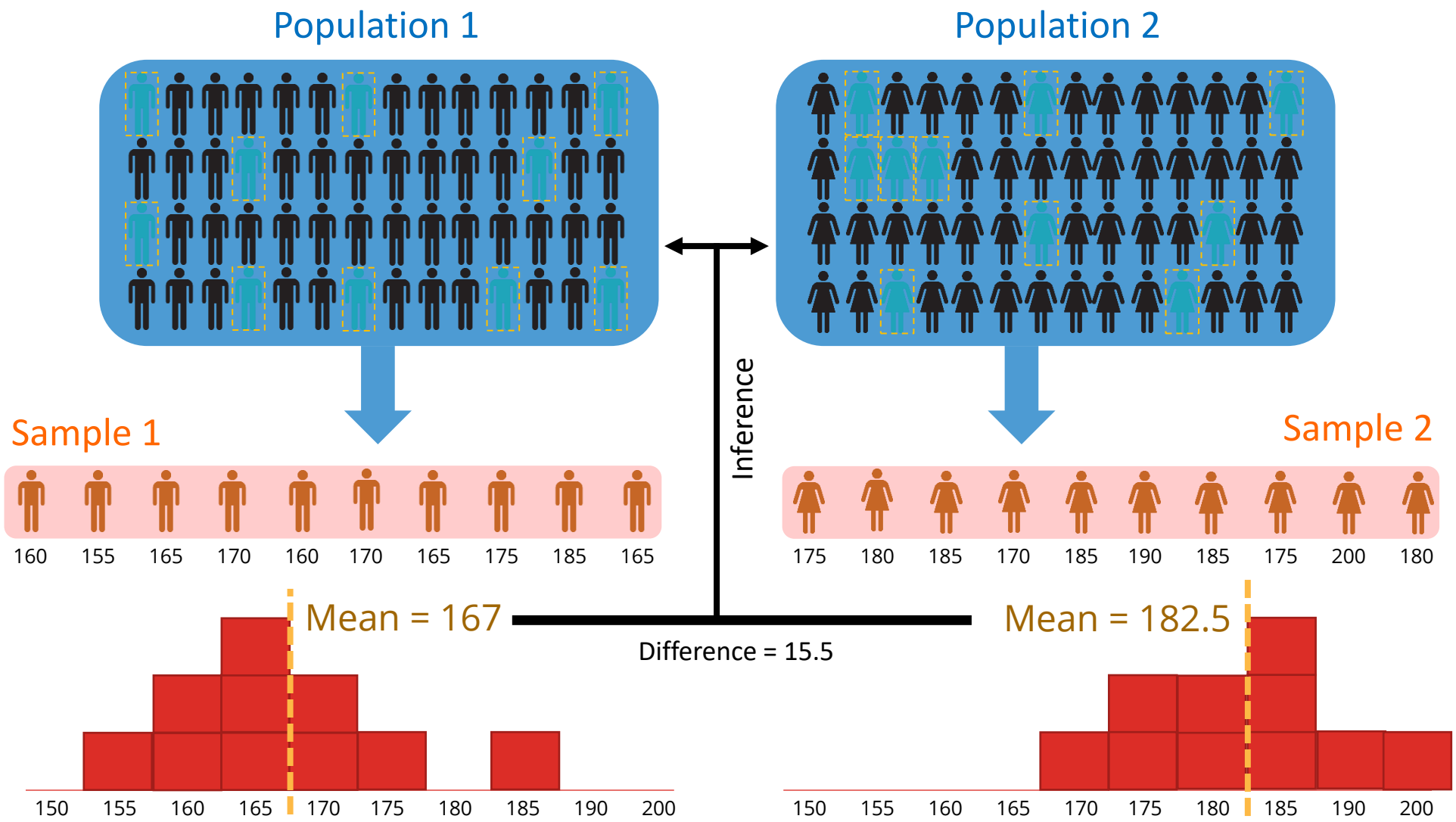
95%CI = 102413.2 to 113586.8

Step 4. Make a conclusion

- Reject H_0
- There is statistically significant evidence that the mean annual income of Australian households differs from $100,000.

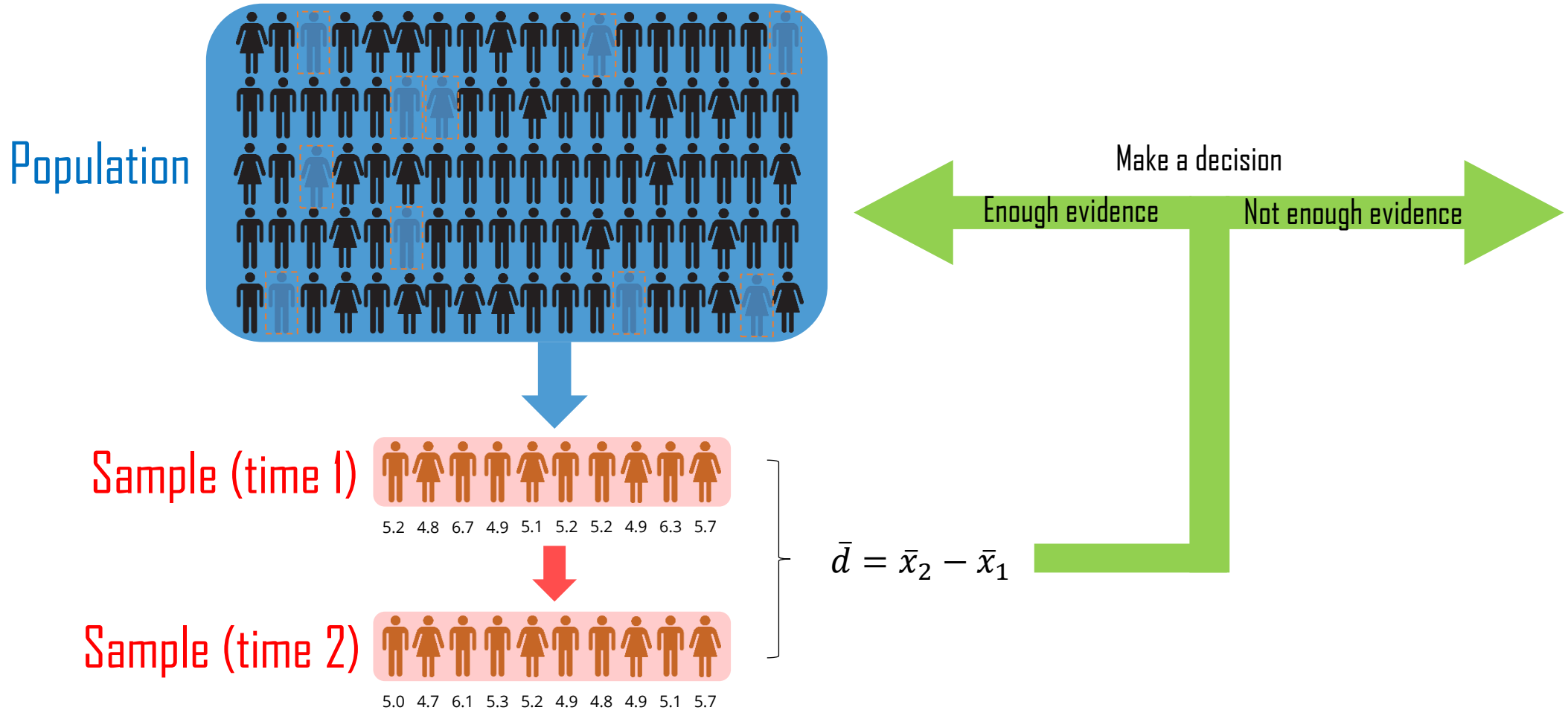

How Many Outcomes?
Outcome Type?
How Many Exp. Vars?
Exp. Var. Type?
If Exp. == Cat, how many levels?
If Exp. == Cat, are levels different or same?
One
Two+
Continuous
Categorical
One
Two+
One
Two+
Continuous
Reference
Categorical
Two
Two+
Same
Different
Same
Different
Continuous
Categorical
Both
Continuous
Categorical
Both
Different
Different
Continuous
Categorical
Both
Different
Different
Categorical
Categorical
Categorical
Correlation / Regression
One sample t-test
Paired sample t-test
Two sample t-test
Repeated measures ANOVA
One way ANOVA
Multiple Regression
Factorial ANOVA / M Regression
Factorial Mixed ANOVA
ANCOVA / M Regression
Logistic Regression
Chi Square
Logistic Regression
Loglinear Regression
Logistic Regression
MANOVA
Factorial MANOVA
MANCOVA

Two Sample t-test



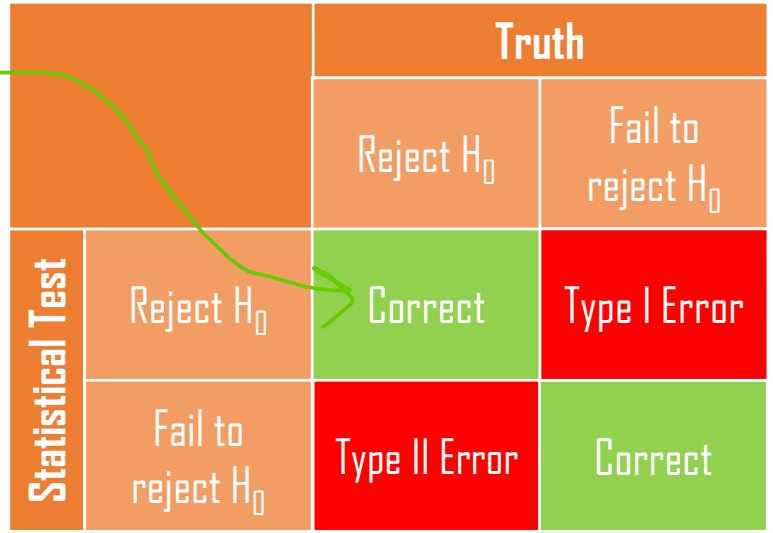
MONASH University

Paired t-test



Type I and Type II Errors

$\text{Power} = 1 - \beta$



$P(\text{Type I Error}) = \alpha$

$P(\text{Type II Error}) = \beta$